

CareerSignal Product Requirements Document (PRD)

1. Document Information

Product Name: CareerSignal

Product Type: Placement Intelligence & Student Readiness Analytics Platform

Version: V1.0

Status: MVP Concept & Analytics Prototype

2. Executive Summary

CareerSignal is a placement intelligence platform designed to help colleges, placement cells, and career counselors improve placement outcomes through data-driven decision making.

Institutions collect large amounts of student, skill, and placement data, but much of it remains underutilized. Placement teams often lack visibility into recruitment funnel performance, branch-level outcomes, student readiness, and employability gaps.

CareerSignal consolidates placement, application, and skill data into a centralized analytics platform that enables stakeholders to identify bottlenecks, benchmark performance, and prioritize interventions that improve placement success.

3. Problem Statement

Most colleges face challenges in answering critical placement-related questions:

- Which branches are underperforming in placements?
- Where do students drop off in the recruitment funnel?
- Which skills are most closely associated with successful placements?
- Which student groups require additional support?
- How effective are current placement preparation efforts?

Without a structured analytics framework, placement decisions are often reactive rather than proactive.

This results in:

- Lower placement rates
 - Inefficient placement preparation programs
 - Delayed student interventions
 - Limited visibility into employability trends
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4. Product Vision

To build a placement intelligence platform that empowers educational institutions to improve student employability through actionable insights, performance monitoring, and data-driven intervention strategies.

5. Target Users

Placement Officers

Goals

- Improve overall placement rate
- Track recruitment funnel performance
- Compare branch-wise outcomes
- Identify placement bottlenecks

Pain Points

- Lack of centralized reporting
 - Difficulty identifying underperforming groups
 - Limited visibility into funnel drop-offs
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Career Counselors

Goals

- Evaluate student readiness
- Identify skill gaps
- Recommend interventions

Pain Points

- Limited access to readiness indicators
 - No standardized framework for student assessment
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Students

Goals

- Understand placement readiness
- Benchmark performance against peers
- Improve employability

Pain Points

- Limited visibility into placement expectations
 - Unclear understanding of skill deficiencies
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6. User Stories

Placement Officer

As a placement officer,

I want to monitor placement rates and funnel performance, so that I can improve placement outcomes across departments.

Career Counselor

As a career counselor,

I want to identify students with low readiness indicators, so that I can provide targeted support and training.

Student

As a student,

I want to understand how my skills compare to successful candidates, so that I can focus on the areas that matter most for placements.

7. Success Metrics

North Star Metric

Placement Rate

Percentage of students successfully placed.

Formula:

Placement Rate = Placed Students / Total Students

Supporting Metrics

Recruitment Funnel

- OA Conversion Rate

- Interview Conversion Rate
- Interview-to-Offer Conversion Rate

Placement Outcomes

- Average Package (LPA)
- Students Placed
- Offer Distribution

Readiness Indicators

- Resume Score
 - Internship Count
 - DSA Score
 - Communication Score
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8. Functional Requirements

FR-1 Executive Overview Dashboard

Objective

Provide a high-level summary of placement performance.

Features

- Placement Rate
 - Average Package
 - Total Students
 - Students Placed
 - Top Performing Branch
 - Branch-wise Placement Analysis
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FR-2 Recruitment Funnel Dashboard

Objective

Analyze candidate progression through the placement pipeline.

Features

- Applications Sent
- OA Attempted
- OA Cleared
- Interviews Received
- Offers Received

Outcomes

- Identify funnel bottlenecks
 - Measure recruitment efficiency
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FR-3 Skill Intelligence Dashboard

Objective

Assess employability and readiness indicators.

Features

- Skill Benchmarking
- Branch Skill Comparison
- Placement Correlation Analysis
- College Tier Comparison

Outcomes

- Identify skill gaps
 - Understand drivers of placement success
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9. Data Model

The platform is built on four interconnected datasets:

Students

Student demographics and academic information.

Skills

Technical and employability skill indicators.

Applications

Recruitment funnel activity and assessment outcomes.

Placements

Placement status, package information, and offer outcomes.

The database follows a relational schema centered around student_id.

[Insert ER Diagram]

10. MVP Scope

Included

- Placement Analytics
 - Funnel Analytics
 - Skill Analytics
 - Branch Comparison
 - College Tier Analysis
 - Interactive Dashboards
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Excluded

- Real-time integrations
 - Student authentication
 - Resume parsing
 - Placement prediction models
 - External job portal integrations
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11. Key Insights from MVP

Placement Performance

- Overall Placement Rate: 45.74%
- Students Placed: 2,287
- Average Package: 6.08 LPA

Recruitment Funnel

- Significant drop-off occurs between OA Attempted and OA Cleared stages.
- Interview-to-Offer conversion remains relatively low, highlighting potential interview readiness gaps.

Skills

- Communication scores remain consistently strong across branches.
 - DSA performance demonstrates a positive relationship with placement outcomes.
 - Skill readiness varies across academic branches.
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12. Future Roadmap

V2 – Readiness Intelligence

Features:

- Student Readiness Index
- Risk Segmentation

- Intervention Recommendations
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V3 – Predictive Analytics

Features:

- Placement Probability Prediction
 - Package Prediction Models
 - Placement Forecasting
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V4 – Career Intelligence Platform

Features:

- Resume Recommendation Engine
 - Personalized Skill Development Plans
 - Skill Gap Analysis
 - Learning Roadmaps
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13. Expected Business Impact

For Placement Teams

- Improved visibility into placement performance
- Faster identification of bottlenecks
- Data-driven decision making

For Career Counselors

- Better student support
- Structured intervention planning

For Students

- Improved placement preparation
- Greater awareness of employability gaps

For Institutions

- Higher placement rates
 - Improved placement outcomes
 - Better recruiter engagement
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14. Conclusion

CareerSignal demonstrates how analytics can transform placement operations from reactive reporting into proactive decision making. By combining placement outcomes, recruitment funnel performance, and skill intelligence into a single platform, institutions can better understand student readiness and improve employability outcomes at scale.